

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES - TECHNICAL CAMPUS
Grade A++ Accredited Institution by NAAC
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An ISO 9001:2015 Certified

SCHOOL OF ENGINEERING & TECHNOLOGY

B.Tech Programme: Artificial Intelligence and Machine Learning – (A) (7th Semester)

Course Title: Predictive Business Analysis Lab

Course Code: AIML409P

Submitted By:
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Enrolment No: 01117711623

VISION OF INSTITUTE

To be an educational institute that empowers the field of engineering to build a sustainable future by providing quality education with innovative practices that supports people, planet and profit.

MISSION OF INSTITUTE

To groom the future engineers by providing value-based education and awakening students' curiosity, nurturing creativity and building capabilities to enable them to make significant contributions to the world.

S. No.	Experiment	Date	Marks (Total Ten) which includes (Laboratory Assessment, Class participation and Viva)	Remarks	Updated Marks	Faculty Signature
1.	Write a program to analysis of a business dataset to identify patterns, correlations, and trends. Visualize the data using techniques such as histograms, scatter plots, and box plots to gain insights into the data distribution and relationships between variables.					
2.	Build a regression model to predict a business metric, such as sales or customer churn, based on relevant features.					
3.	Write a program to apply different regression algorithms, such as linear regression, decision trees, or random forests, and compare their performance using evaluation metrics like Mean Squared Error (MSE) or R-squared.					
4.	Perform a classification analysis to predict a binary or multi-class business outcome, such as customer segmentation or product categorization. Experiment with classification algorithms like logistic regression, support vector machines (SVM), or decision trees. Evaluate the models using metrics such as accuracy, precision, recall, and F1-score.					

5.	To apply time series forecasting techniques to predict future business trends or product/service demand.					
6.	To investigate the impact of feature selection and feature engineering techniques on predictive modeling performance.					
7.	Write a program to experiment with different methods such as correlation analysis, stepwise regression, or Recursive Feature Elimination (RFE) to identify the most relevant features. Compare the performance of models with and without feature engineering.					
8.	Write a program to explore the effectiveness of ensemble methods, such as bagging, boosting, or stacking, in improving predictive performance.					
9.	Write a program to Analyze textual data, such as customer reviews or social media comments, to extract sentiment and gain insights into customer opinions. Apply techniques like Natural Language Processing (NLP), text preprocessing, and sentiment analysis algorithms to classify text into positive, negative, or neutral sentiments.					
10.	Write a program to analyze the resulting customer segments based on demographic, behavioral, or transactional characteristics to identify					

	distinct customer groups and target them with tailored marketing strategies.					
11.	Write a program to design an A/B test to evaluate the impact of a business intervention or change on key metrics.					
12.	Simulate different business scenarios, such as pricing strategies, inventory optimization, or resource allocation, using predictive models.					